

QuarkShield.ai & DigiCert

Comparative Architecture & Complementary Post-Quantum Services

Author: QuarkShield.ai Strategy & Engineering

Target: DigiCert ONE & Quantum Central Integration

Classification: Confidential / Partner Briefing

Standards: NIST FIPS 203/204/205 • CycloneDX 1.6

Strategic Overview: This analysis explores how QuarkShield.ai's endpoint, host, and codebase cryptographic discovery platform acts as an ideal **complementary telemetry feeder and remediation engine** for the DigiCert ONE and DigiCert Quantum Central platforms.

1. Executive Summary

As global organizations face regulatory mandates (NIST, OMB M-23-02, CNSA 2.0, DORA, PCI-DSS 4.0) to migrate from legacy public-key cryptography (RSA, ECC, Diffie-Hellman) to Post-Quantum Cryptography (PQC), enterprise security leaders require both **PKI issuance/lifecycle management** and **deep cryptographic observability**.

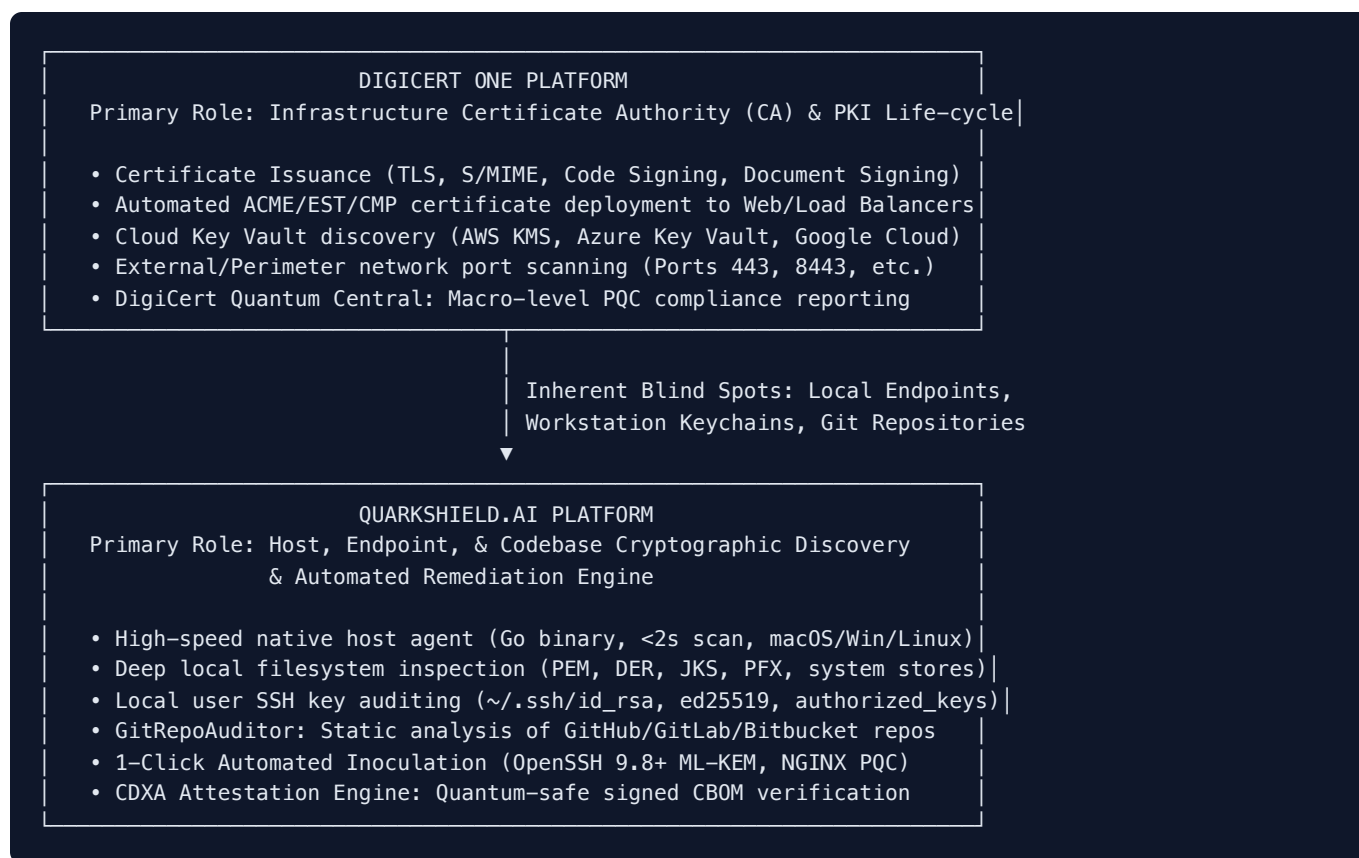
DigiCert is the undisputed market leader in Certificate Authority (CA) infrastructure, issuing public/private certificates and automating web/load balancer lifecycle management. In late 2024 / 2025, DigiCert launched **DigiCert Quantum Central** to track quantum readiness.

However, DigiCert's discovery architecture is inherently **network-centric and cloud-vault-centric**. It lacks visibility into local developer endpoints, workstation keychains, local keystores, and pre-build source code repositories. **QuarkShield.ai fills this exact gap**, providing the host agent and code scanner that feeds comprehensive CycloneDX 1.6 Cryptographic Bills of Materials (CBOMs) into DigiCert.

2. What We Have in Common

DIMENSION	DIGICERT (DIGICERT ONE / QUANTUM CENTRAL)	QUARKSHIELD.AI
Core Mission	Safeguard digital trust against quantum threats ("Harvest Now, Decrypt Later").	Discover, inventory, and remediate quantum vulnerabilities across enterprise fleets.
PQC Standards	Implements NIST standards: FIPS 203 (ML-KEM) , FIPS 204 (ML-DSA) , FIPS 205 (SLH-DSA) .	Audits all assets against NIST FIPS 203/204/205 and CNSA 2.0 quantum resistance levels.
CBOM Standard	Ingests and aggregates Cryptographic Bills of Materials (CBOM / xBOM) in Quantum Central.	Natively generates and exports standardized CycloneDX 1.6 CBOMs with full crypto properties.
Risk Modeling	Evaluates deprecated classical cryptography vulnerable to Shor's algorithm (RSA, ECC).	Computes quantum vulnerability scores (0–100) per machine, code repo, and cryptographic primitive.
Target Market	Global enterprises, MSPs/MSSPs, financial institutions, federal defense, critical infrastructure.	Enterprise CISOs, MSPs, DevSecOps teams, infrastructure architects, and compliance officers.

3. Architectural Roles & Division of Responsibilities



4. DigiCert's Inherent Blind Spots

- **Local Endpoint & Workstation "Shadow Crypto":** Network scanners cannot inspect internal developer MacBooks, Windows workstations, or build runners. Local user SSH keys (~/.ssh/id_rsa), Java KeyStores (.jks), local client certificates, and VPN tunnel configs remain completely dark to DigiCert.
- **Pre-Build Source Code Cryptographic Debt:** DigiCert *Software Trust Manager* signs compiled binaries, but cannot scan source code repositories prior to compilation for hardcoded classical algorithms (e.g. RSA-2048 encryptors, deprecated TLS cipher suites).
- **Host-Level Configuration Remediation:** While DigiCert can *issue* a post-quantum certificate, it does not reconfigure host daemons (e.g., configuring OpenSSH 9.8+ for mlkem768x25519 hybrid key exchange or patching NGINX cipher suites).

5. Four High-Value Services QuarkShield Can Provide to DigiCert

SERVICE 1 • TELEMETRY FEEDER

Host & Desktop "Shadow Crypto" Discovery Feed for DigiCert Quantum Central

The Gap: DigiCert Quantum Central requires a unified cryptographic inventory but relies on third-party CBOM/xBOM ingest for unmanaged host infrastructure and developer workstations.

QuarkShield Solution: QuarkShield's native Go agent (<2s scan) discovers local certificates, private keys, SSH keypairs, and keystores across macOS, Windows, and Linux. QuarkShield automatically packages this into standardized **CycloneDX 1.6 CBOM** format and streams it directly into DigiCert Quantum Central via REST APIs, eliminating enterprise blind spots.

SERVICE 2 • DEVSECOPS SHIFT-LEFT

Git Codebase Cryptographic Auditor for DigiCert Software Trust Manager

The Gap: DigiCert Software Trust Manager signs release binaries in CI/CD, but lacks static analysis to detect cryptographic vulnerabilities in raw source code before build time.

QuarkShield Solution: QuarkShield's `GitRepoAuditor` connects to GitHub, GitLab, and Bitbucket to scan codebases, dependencies (e.g., BouncyCastle, Web3.js), and infrastructure-as-code manifests. It generates a pre-build CBOM, allowing DigiCert's pipeline to enforce PQC policy gating before code signing occurs.

SERVICE 3 • ACTIVE CONFIGURATION

Automated Host & Endpoint Inoculation / Configuration Remediation Playbooks

The Gap: Issuing a PQC certificate does not automatically reconfigure servers, daemons, and client workstations to support post-quantum key exchange or signature schemes.

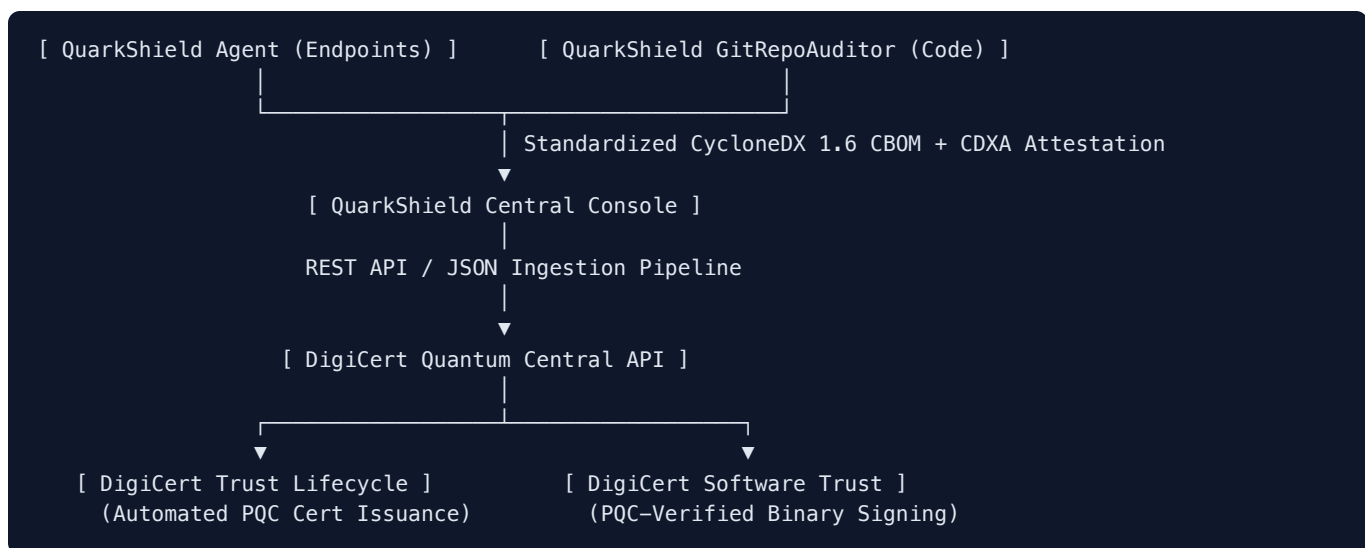
QuarkShield Solution: QuarkShield delivers executable 1-click remediation scripts and Ansible/Intune playbooks that upgrade OpenSSH to hybrid post-quantum key exchange (`snt.rup761x25519-sha512@openssh.com` / `mlkem768x25519`), update NGINX/Apache TLS configs, and generate compliant PQC keypairs.

CDXA Cryptographic Bill of Materials Attestation & Verification Engine

The Gap: Regulators (NIST, OMB M-23-02, CNSA 2.0, DORA) require cryptographically verifiable, tamper-evident proof that an enterprise's cryptographic inventory has been validated.

QuarkShield Solution: QuarkShield signs generated CycloneDX 1.6 CBOMs with **CDXA (Cryptographic Dependency & X.509 Attestation)** using NIST FIPS 204 (ML-DSA-65) post-quantum digital signatures. This verifiable proof can be archived directly in DigiCert's compliance portal.

6. Integration Architecture & Data Flow



7. Strategic Partnership & Business Value

- **Accelerates DigiCert PQC Certificate Sales:** By uncovering thousands of vulnerable RSA/ECC keys across enterprise workstations and hosts, QuarkShield creates immediate commercial demand for DigiCert's new post-quantum certificates and Trust Lifecycle Manager subscriptions.
- **360-Degree Visibility for DigiCert Customers:** Expands DigiCert's coverage beyond perimeter web servers to developer workstations, internal servers, and source code.
- **Seamless CycloneDX 1.6 Interoperability:** Zero proprietary lock-in. QuarkShield's native CycloneDX 1.6 output maps directly into DigiCert ONE schemas without complex transformations.